Automatic Differentiation

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Differentiation approaches

We can compute partial derivatives in different ways:

1. **Symbolically**, by fixing one of the variables and differentiating with respect to the others, either manually or using a computer.
2. **Numerically**, using the formula $f'(x) \approx (f(x+h) - f(x))/h$.
3. **Algorithmically**, either forward or reverse : this is what we will explore here.

Chain rule

Consider $f(x) = f_3(f_2(f_1(x)))$. If we don't have the expression of f_1 but we can only evaluate $f_i(x)$ or $f'(x)$ for a given x ? The chain rule gives

$$f'(x) = f_3'(f_2(f_1(x))) \cdot f_2'(f_1(x)) \cdot f_1'(x).$$

Let's define $s_0 = x$ and $s_k = f_k(s_{k-1})$, we now have:

$$f'(x) = f_3'(s_2) \cdot f_2'(s_1) \cdot f_1'(s_0).$$

Two choices here:

Forward	Reverse
$t_0 = 1$	$r_3 = 1$
$t_k = f_k'(s_{k-1}) \cdot t_{k-1}$	$r_k = r_{k+1} \cdot f_{k+1}'(s_k)$

Forward Differentiation

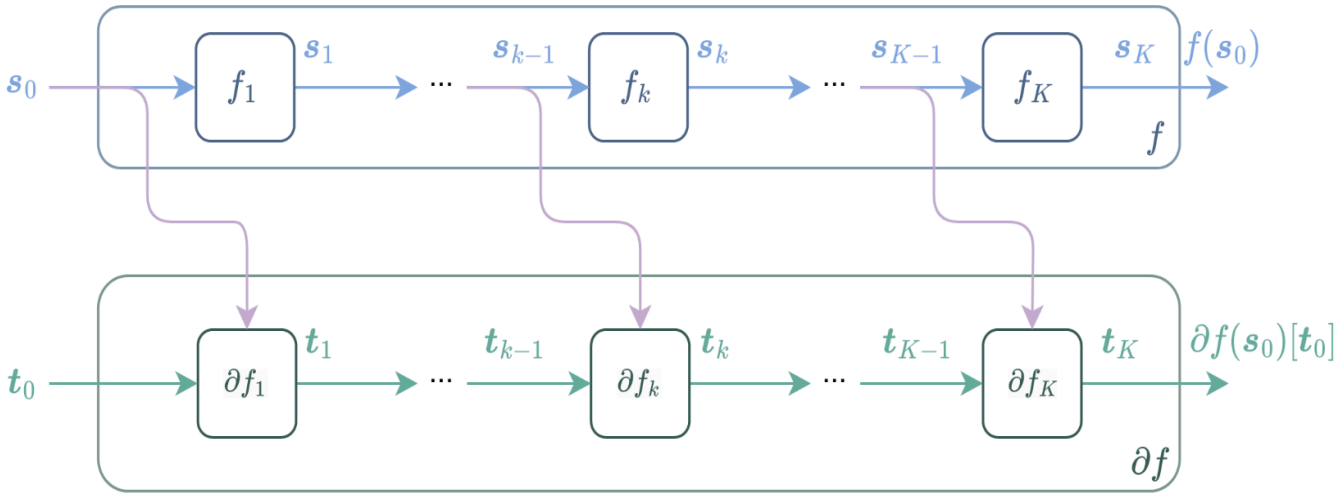


Figure 8.1

Implementation

```
struct Dual{T}
    value::T # s_k
    derivative::T # t_k
end
```

```
Base.:-(x::Dual{T}) where {T} = Dual(-x.value, -x.derivative)
```

```
Base.:*(x::Dual{T}, y::Dual{T}) where {T} = Dual(x.value * y.value, x.value * y.derivative + x.derivative * y.value)
```

```
Dual(-3, -10)
```

```
-Dual(1, 2) * Dual(3, 4)
```

```
f_1 (generic function with 1 method)
```

```
f_1(x, y) = x * y
```

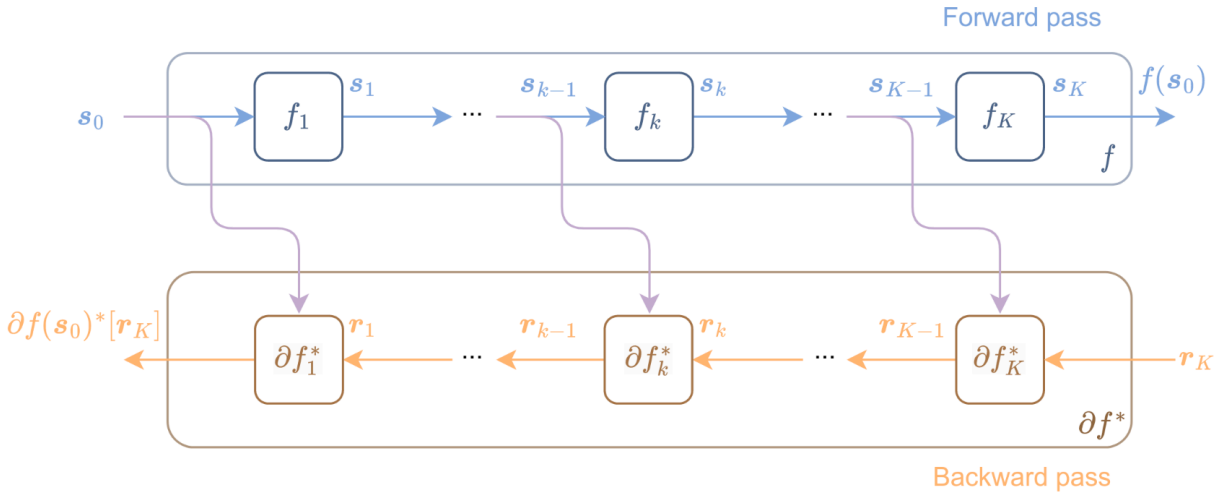
```
f_2 (generic function with 1 method)
```

```
f_2(s1) = -s1
```

```
Dual(-3, -10)
```

```
(f_2 ∘ f_1)(Dual(1, 2), Dual(3, 4))
```

Reverse Differentiation



Two different takes on the multivariate chain rule

The chain rule gives us

$$\frac{\partial f_3}{\partial x}(f_1(x), f_2(x)) = \partial_1 f_3(s_1, s_2) \cdot \frac{\partial s_1}{\partial x} + \partial_2 f_3(s_1, s_2) \cdot \frac{\partial s_2}{\partial x}$$

To compute this expression, we need the values of $s_1(x)$ and $s_2(x)$ as well as the derivatives $\partial s_1/\partial x$ and $\partial s_2/\partial x$.

Common to forward and reverse: Given s_1, s_2, computes **local** derivatives $\partial_1 f_3(s_1, s_2)$ and $\partial_2 f_3(s_1, s_2)$, shortened $\partial_1 f_3, \partial_2 f_3$ for conciseness.

Forward

$$\begin{aligned} t_3 &= \partial_1 f_3 \cdot t_1 + \partial_2 f_3 \cdot t_2 \\ &= \begin{bmatrix} \partial_1 f_3 & \partial_2 f_3 \end{bmatrix} \cdot \begin{bmatrix} t_1 \\ t_2 \end{bmatrix} \\ &= \partial f_3 \cdot \begin{bmatrix} t_1 \\ t_2 \end{bmatrix} \end{aligned}$$

Reverse

$$\begin{aligned} \begin{bmatrix} r_1 & r_2 \end{bmatrix} &\mathrel{+}= r_3 \cdot \partial f_3 \\ &\mathrel{+}= r_3 \cdot \begin{bmatrix} \partial_1 f_3 & \partial_2 f_3 \end{bmatrix} \\ &\mathrel{+}= \begin{bmatrix} r_3 \cdot \partial_1 f_3 & r_3 \cdot \partial_2 f_3 \end{bmatrix} \end{aligned}$$

Reverse*

$$\begin{aligned} \begin{bmatrix} r_1 \\ r_2 \end{bmatrix} &\mathrel{+}= \partial f_3^* \cdot r_3 \\ &\mathrel{+}= \begin{bmatrix} \partial_1 f_3 \\ \partial_2 f_3 \end{bmatrix} \cdot r_3 \\ &\mathrel{+}= \begin{bmatrix} \partial_1 f_3 \cdot r_3 \\ \partial_2 f_3 \cdot r_3 \end{bmatrix} \end{aligned}$$

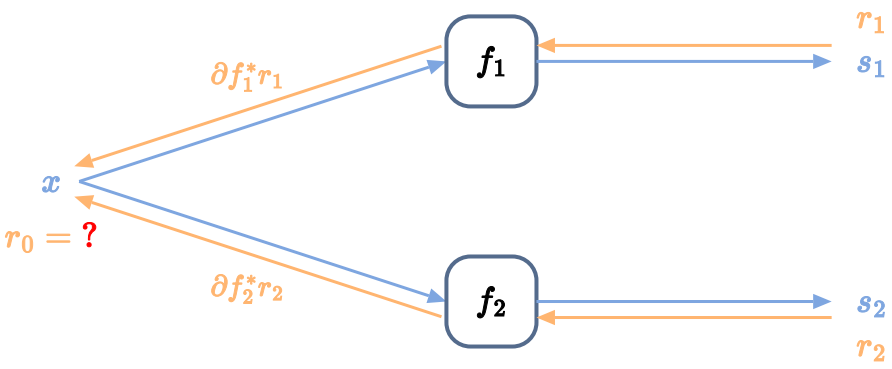
When using automatic differentiation, don't forget that we must always evaluate the derivatives. For the following example we choose to evaluate it in $x = 3$

▸ **Apply the automatic differentiation to $s_3 = f_3(s_1, s_2) = s_1 + s_2$, with $s_1 = f_1(x) = x$ and $s_2 = f_2(x) = x^2$**

Reverse mode on a DAG

- Forward mode: we know how to propagate t through any graph.
- Reverse mode on a chain: we know that $r_{k-1} = \partial f_k^* r_k$.
- Reverse mode on a DAG: ❓

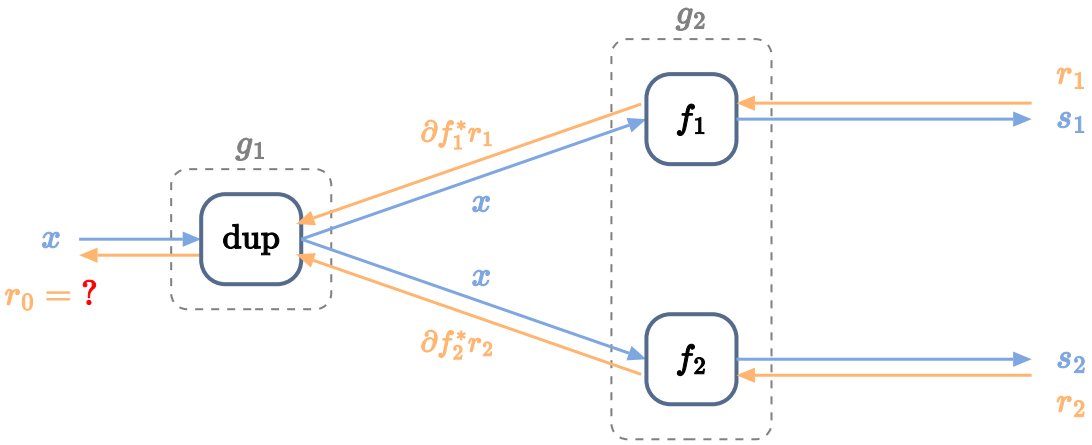
Consider the simplest DAG that is not a chain: x is used by both f_1 and f_2. The reverse pass brings two contributions $\partial f_1^* r_1$ and $\partial f_2^* r_2$ back to the same x.



▸ **How should we combine $\partial f_1^* r_1$ and $\partial f_2^* r_2$ into r_0 ? Sum them ? Multiply them ?**

Let's turn it into a chain!

Key idea: introduce $\mathbf{dup}(x) = (x, x)$ so that $f = g_2 \circ g_1$ is a chain with $g_1 = \mathbf{dup}$ and $g_2(x_1, x_2) = (f_1(x_1), f_2(x_2))$.



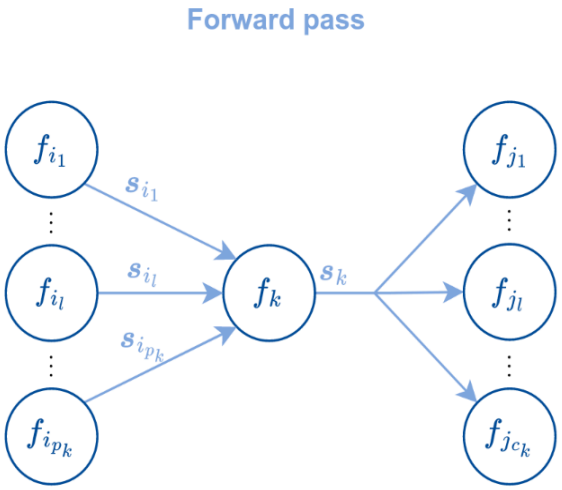
f_1 and f_2 now contribute to **different** entries of the output of **dup** so there is nothing to combine:

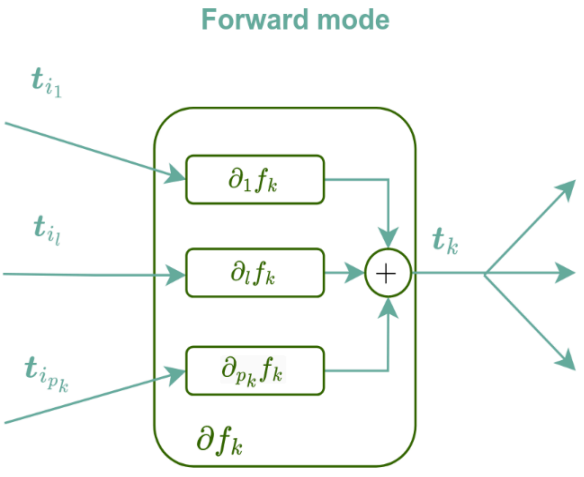
$$\partial g_2^* \begin{bmatrix} r_1 \\ r_2 \end{bmatrix} = \begin{bmatrix} \partial f_1^* & 0 \\ 0 & \partial f_2^* \end{bmatrix} \begin{bmatrix} r_1 \\ r_2 \end{bmatrix} = \begin{bmatrix} \partial f_1^* r_1 \\ \partial f_2^* r_2 \end{bmatrix}$$

What remains is the reverse pass of **dup**... but **dup** is a function like any other: we can compute its Jacobian!

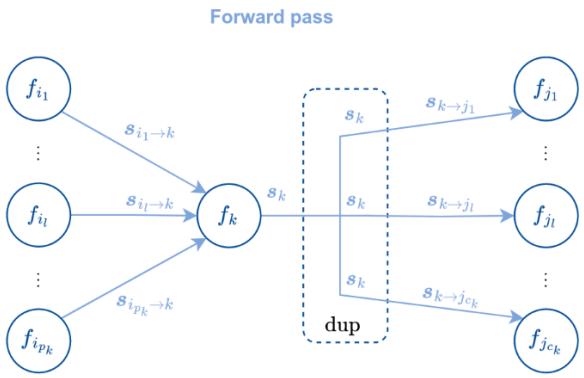
▸ **What is the reverse pass of dup ?**

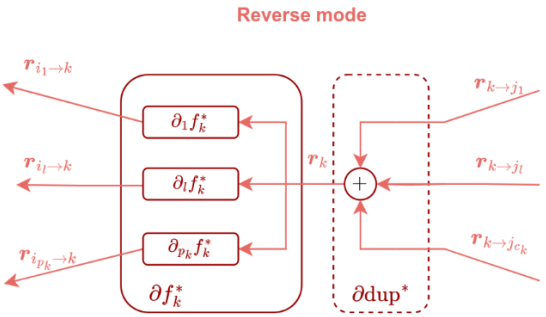
Forward tangents





Reverse tangents





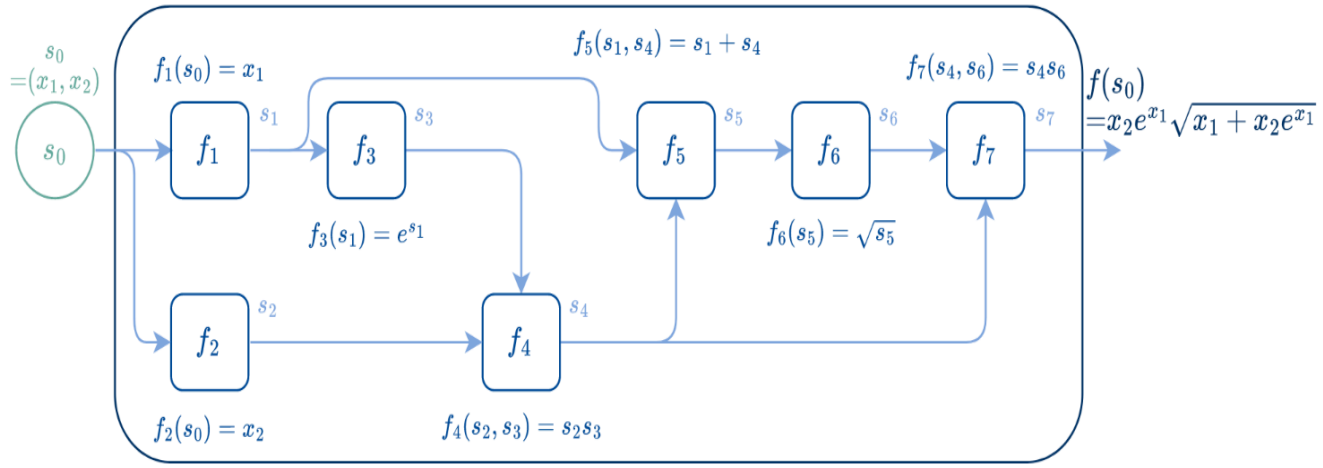
▶ **Why is ∂dup^* a sum ?**

Computation graph

We build the graph of

$$f(x_1, x_2) = x_2 e^{x_1} \sqrt{x_1 + x_2 e^{x_2}}$$

at $x_1 = 1, x_2 = 2$. The slider first reveals the forward evaluation of each node v_i, then accumulates the adjoints $\bar{v}_i = \partial f / \partial v_i$ backwards, one node at a time.



▶ **Can this directed graph have cycles ?**

▶ **What happens if f_4 is handled before f_5 in the backward pass ?**

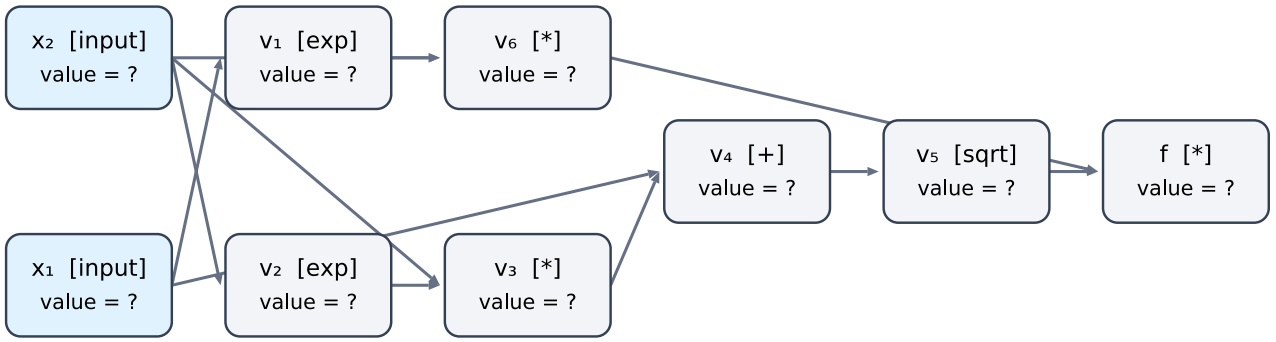
▶ **How to prevent this from happening ?**

Reverse mode in action

$$f(x_1, x_2) = x_2 e^{x_1} \sqrt{x_1 + x_2 e^{x_2}}$$

◀ ▶ 1

Computation graph



Comparison

- Forward mode of $f(x)$ with dual numbers `Dual.(x, v)` computes Jacobian-Vector Product (JVP) $J_f(x) \cdot v$
- Reverse mode of $f(x)$ computes Vector-Jacobian Product (VJP) $v^\top J_f(x)$ or in other words $J_v(x)^\top v$

▸ **How can we compute the full Jacobian ?**

▸ **When is each mode faster than the other one to compute the full Jacobian ?**

▸ **When is the speed of numerical differentation comparable to autodiff ?**

From forward to reverse mode

Consider $f(x_1, x_2) = x_1^2 + x_2^2$ at $s_0 = (1, 2)$. We decompose it as $f = f_2 \circ f_1$ and compute the **local** Jacobians:

$$
\begin{aligned}
f_1(x_1, x_2) &= \begin{bmatrix} x_1^2 \\ x_2^2 \end{bmatrix} & s_1 = f_1(s_0) &= \begin{bmatrix} 1 \\ 4 \end{bmatrix} & \partial f_1(s_0) &= \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix} \\
f_2(x_1, x_2) &= x_1 + x_2 & s_2 = f_2(s_1) &= 5 & \partial f_2(s_1) &= \begin{bmatrix} 1 & 1 \end{bmatrix}
\end{aligned}
$$

Forward mode computes one column of the Jacobian $\partial f(s_0)$ for each one-hot tangent t_0:

$$
\begin{aligned}
t_0 &= \begin{bmatrix} 1 \\ 0 \end{bmatrix} & t_1 = \partial f_1(s_0) \cdot t_0 &= \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix} & t_2 = \partial f_2(s_1) \cdot t_1 &= \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \end{bmatrix} = 2 \\
t_0 &= \begin{bmatrix} 0 \\ 1 \end{bmatrix} & t_1 = \partial f_1(s_0) \cdot t_0 &= \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 4 \end{bmatrix} & t_2 = \partial f_2(s_1) \cdot t_1 &= \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 4 \end{bmatrix} = 4
\end{aligned}
$$

From forward to reverse mode

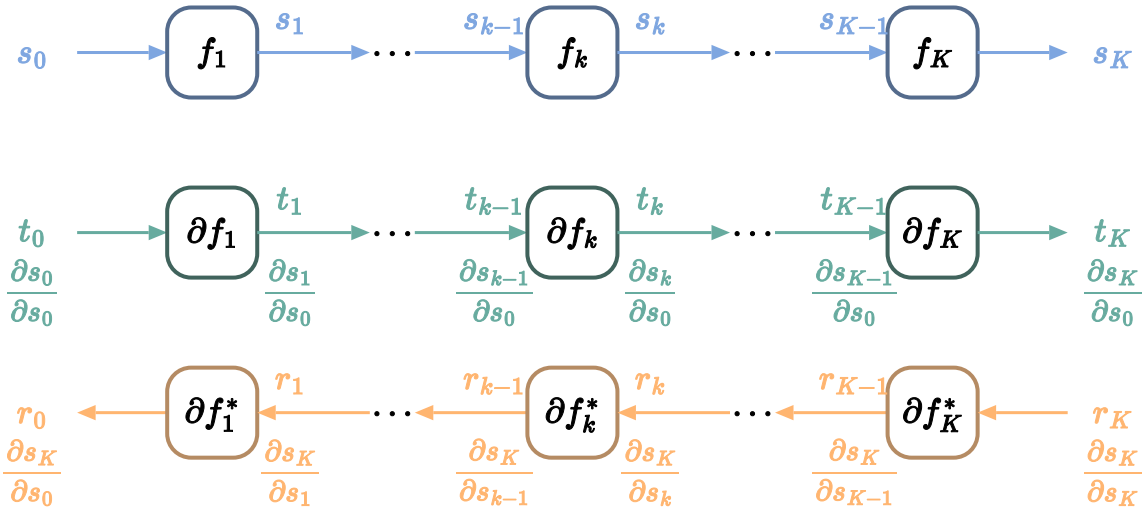
Concatenating the two columns gives the full Jacobian. We can then factor $\partial f_2(s_1)$ and then $\partial f_1(s_0)$ out of the concatenation:

$$
\begin{aligned}
\partial f(s_0) &= \begin{bmatrix} 2 & 4 \end{bmatrix} \\
&= \begin{bmatrix} \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} & \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \end{bmatrix} \\
&= \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} & \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \end{bmatrix} \\
&= \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} & \begin{bmatrix} 0 \\ 1 \end{bmatrix} \end{bmatrix} \\
&= \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix} \cancel{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}}
\end{aligned}
$$

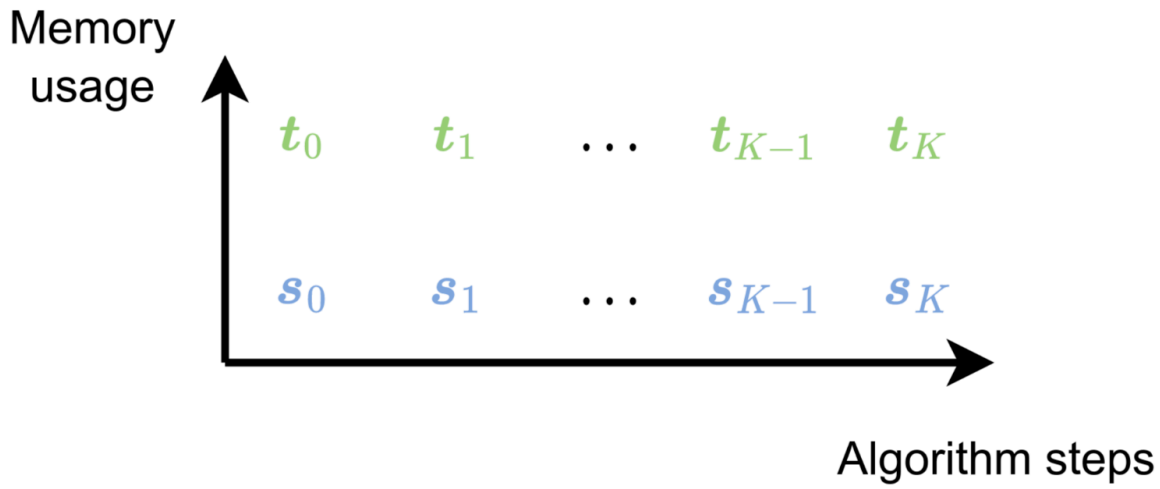
The concatenation of the one-hot tangents is the identity so it disappears! We are left with $\partial f_2(s_1) \cdot \partial f_1(s_0)$ that we can compute from left to right in a single pass: this is reverse mode!

$$
r_2 = 1 \qquad r_1 = r_2 \cdot \partial f_2(s_1) = \begin{bmatrix} 1 & 1 \end{bmatrix} \qquad r_0 = r_1 \cdot \partial f_1(s_0) = \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix} = \begin{bmatrix} 2 & 4 \end{bmatrix}
$$

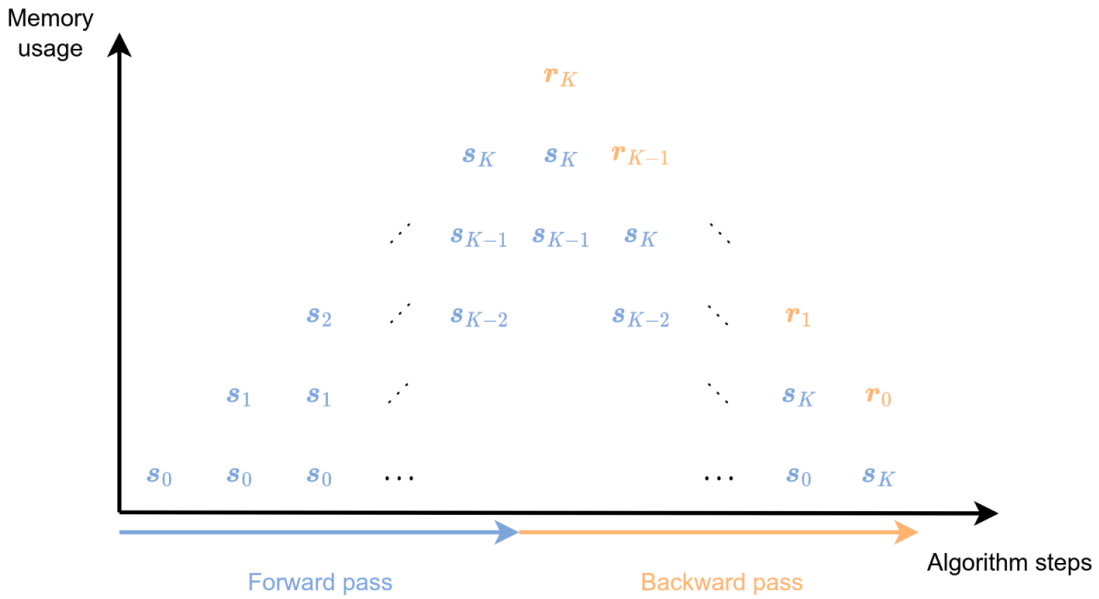
Interpretation of tangents



Memory usage of forward mode



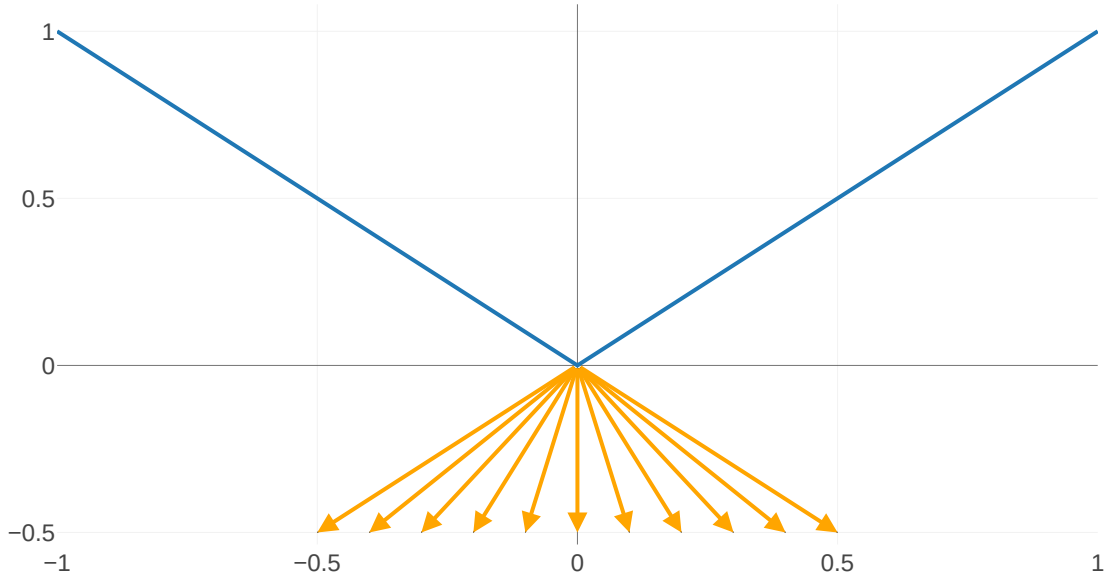
Memory usage of reverse mode



▶ **Why don't we show the local Jacobian ∂J_k in the memory usage ?**

Discontinuity

▶ **Is the function $|x|$ is differentiable at $x = 0$?.**



▶ **What about returning a convex combination of the derivative from the left and right ?**

Forward mode

```
abs (generic function with 1 method)
```

```
1 abs(x) = ifelse(x < 0, -x, x)
```

```
abs_bis (generic function with 1 method)
```

```
1 abs_bis(x) = ifelse(x > 0, x, -x)
```

```
Base.isless(x::Dual, y::Real) = isless(x.value, y)
```

```
Base.isless(x::Real, y::Dual) = isless(x, y.value)
```

▸ Dual(0, 1)

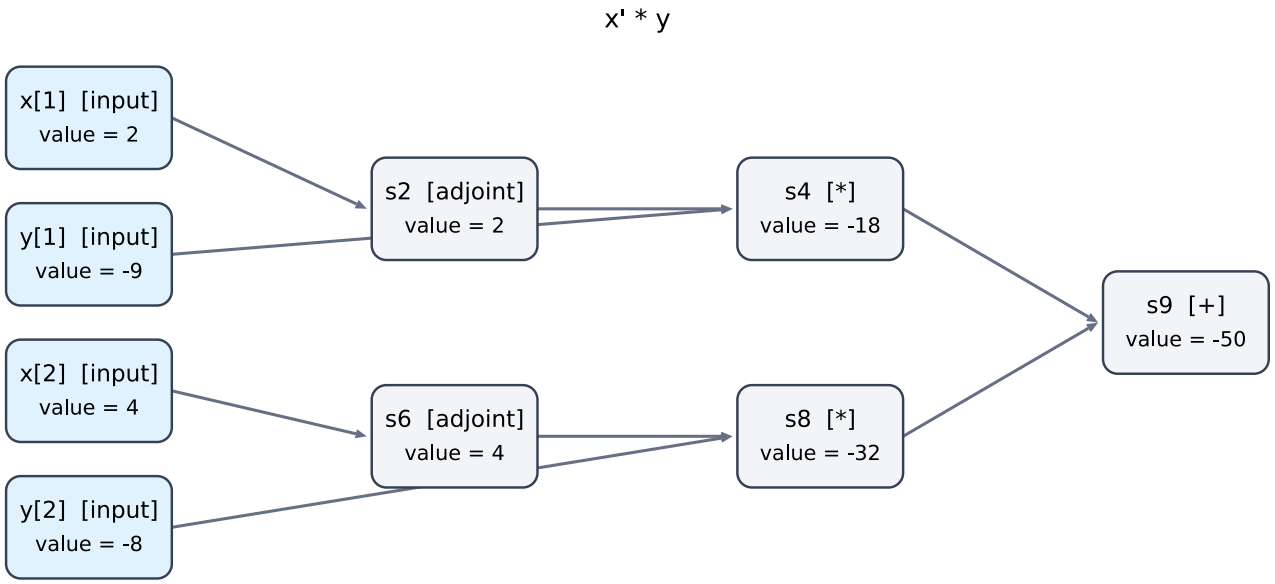
```
abs(Dual(0, 1))
```

▸ Dual(0, -1)

```
abs_bis(Dual(0, 1))
```

Issues with scalar AD

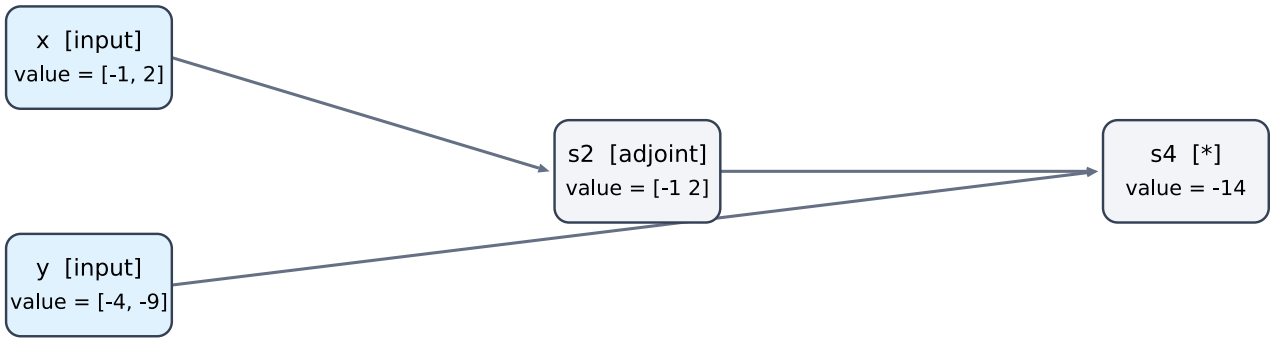
n = 2



Nodes with vector value

n = 2

x' * y



Disadvantages of scalar-valued nodes

- If AD in an interpreted language like Python, the bottleneck of scalar-valued AD becomes the Python interpreter. Solved by using compiled language like Julia/Rust/C/C++/...
- If each node is heap allocated (e.g. with `malloc` in C), not cache friendly. Solved by a centralized storage for all the values of all nodes, e.g. `MathOptInterface.Nonlinear.ReverseAD`'s values vector (TODO hyperlink)
- Not using optimized BLAS routines or GPU kernels for matrix-matrix multiplications. Not solveable for scalar-valued AD, need vector-valued AD!

Neural network

1. Start with an input vector $\boldsymbol{x}$.
2. Apply a matrix product $\boldsymbol{y} = \boldsymbol{W_1 x}$ (or $\boldsymbol{y} = \boldsymbol{W_1 x} + \boldsymbol{b_1}$ with bias).
3. Apply an activation function, say `tanh`, independently to each entry: $\boldsymbol{z_i} = \mathbf{tanh}(\boldsymbol{y_i})$. Using the broadcasting syntax of Julia, we can write $\boldsymbol{z} = \mathbf{tanh.}\,(\boldsymbol{y})$. Usually denoted $\boldsymbol{\sigma(y)}$.
4. Repeat with $\boldsymbol{W_2}$ for next layer etc...

$$\hat{\boldsymbol{y}} = \boldsymbol{W_\ell}\,\mathbf{tanh.}\,(\boldsymbol{W_{\ell-1}}\,\mathbf{tanh.}\,(\cdots \boldsymbol{W_2}\,\mathbf{tanh.}\,(\boldsymbol{W_1 x})))$$

Batching

Applying a neural network to 2 inputs can be done simultaneously so that **matrix-vector** products become **matrix-matrix** product, which better exploit the parallelization capabilities of CPUs and GPUs. Let $\boldsymbol{X} = \begin{bmatrix}\boldsymbol{x^{(1)}} & \boldsymbol{x^{(2)}}\end{bmatrix}$:

$$\begin{aligned}\begin{bmatrix}\sigma(\boldsymbol{W_1 x^{(1)}}) & \sigma(\boldsymbol{W_1 x^{(2)}})\end{bmatrix} &= \sigma(\begin{bmatrix}\boldsymbol{W_1 x^{(1)}} & \boldsymbol{W_1 x^{(2)}}\end{bmatrix}) \\ &= \sigma(\boldsymbol{W_1}\begin{bmatrix}\boldsymbol{x^{(1)}} & \boldsymbol{x^{(2)}}\end{bmatrix}) \\ &= \sigma(\boldsymbol{W_1 X})\end{aligned}$$

With bias, let $\mathbf{1}^\top = \begin{bmatrix}1 & 1\end{bmatrix}$:

$$\begin{bmatrix}\boldsymbol{W_1 x^{(1)}} + \boldsymbol{b_1} & \boldsymbol{W_1 x^{(2)}} + \boldsymbol{b_1}\end{bmatrix} = \begin{bmatrix}\boldsymbol{W_1 x^{(1)}} & \boldsymbol{W_1 x^{(2)}}\end{bmatrix} + \begin{bmatrix}\boldsymbol{b_1} & \boldsymbol{b_1}\end{bmatrix} = \boldsymbol{W_1 X} + \boldsymbol{b_1}\mathbf{1}^\top$$

Two equivalent approaches

$\boldsymbol{b_k}$ is a **column** vector, $\boldsymbol{S_i}, \boldsymbol{X}, \boldsymbol{W_i}, \boldsymbol{Y}$ are matrices.

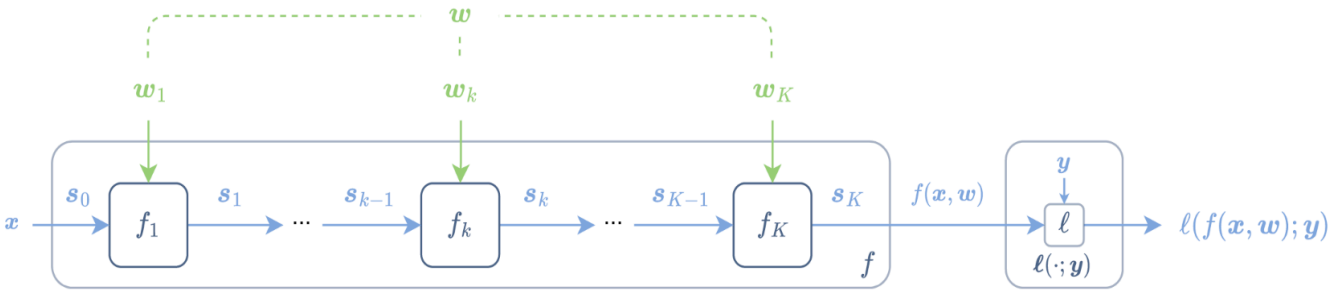
Right-to-left

$$
\begin{aligned}
S_0 &= X \\
S_{2k-1} &= W_k S_{2k-2} + b_k \mathbf{1}^\top \\
S_{2k} &= \sigma(S_{2k-1}) \\
S_{2H+1} &= W_{k+1} S_{2H} \\
S_{2H+2} &= \ell(S_{2H+1}; Y)
\end{aligned}
$$

Left-to-right

$$
\begin{aligned}
S_0 &= X \\
S_{2k-1} &= S_{2k-2} W_k + \mathbf{1} b_k^\top \\
S_{2k} &= \sigma(S_{2k-1}) \\
S_{2H+1} &= S_{2H} W_{k+1} \\
S_{2H+2} &= \ell(S_{2H+1}; Y)
\end{aligned}
$$

Evaluation



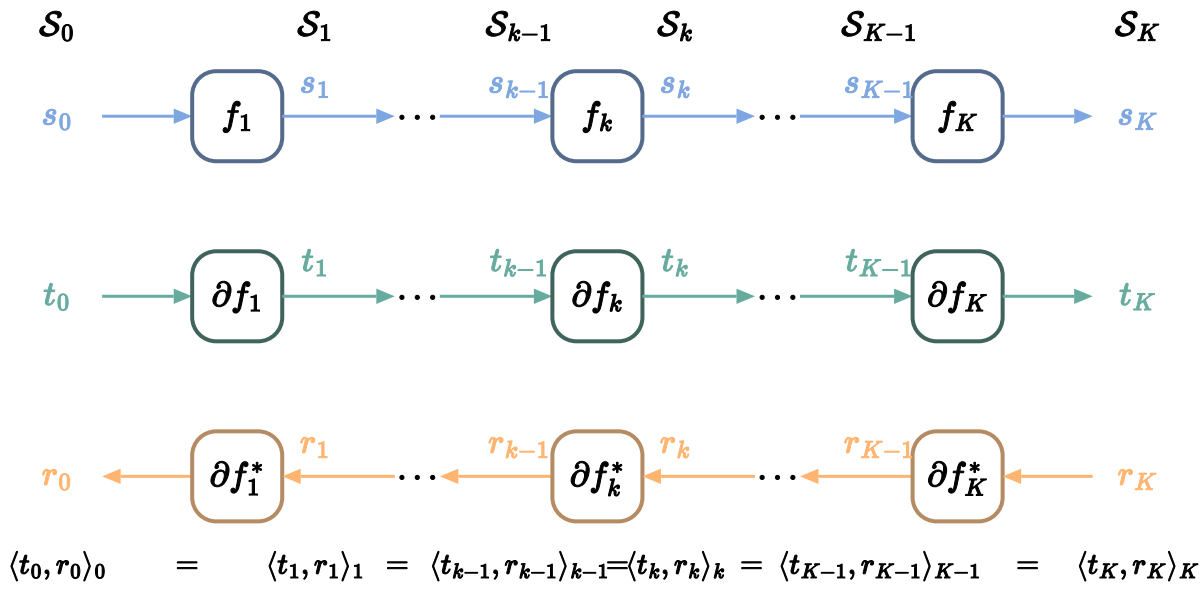
Spaces and scalar products

Let $(\mathcal{S}_k)_k$ be the spaces for which $s_k, t_k, r_k \in \mathcal{S}_k$ and $f_k : \mathcal{S}_{k-1} \to \mathcal{S}_k$.

In the vectorized case, the Jacobian ∂f_k may be expressed more naturally as a linear map $\partial f_k : \mathcal{S}_{k-1} \to \mathcal{S}_k$ and $t_k = \partial f_k(t_{k-1})$. To obtain the matrix-form of the Jacobian, we need to fix a basis for $\mathcal{S}_{k-1}$ and $\mathcal{S}_k$ and vectorize t_k and t_{k-1} according to that basis which is often cumbersome.

Instead, we choose a scalar products $\langle \cdot, \cdot \rangle_k$ for $\mathcal{S}_k$ and the **adjoint** of the Jacobians according to these bases in the backward pass. For instance, if $\mathcal{S}_K = \mathcal{R}$, $t_K = \langle t_K, 1 \rangle_K$ for the canonical scalar product of $\mathcal{R}$ so we pick $r_K = 1$. We then search for a way to obtain $r_0(r_K)$ so that $\langle t_0, r_0(r_K) \rangle_0 = \langle \partial f_K(\cdots \partial f_1(t_0), r_K \rangle_K$. Then, for instance if $\mathcal{S}_0 = \mathbb{R}^n$, with the canonical scalar product $\langle x, y \rangle_0 = x^\top y$, we have $\partial_i f(s_0) = \langle e_i, r_0(r_K) \rangle_0 = (r_0(r_K))_i$.

Spaces and scalar products



Adjoint

The *adjoint* of a linear map $A : \mathcal{S}_1 \to \mathcal{S}_2$ for scalar products $\langle \cdot, \cdot \rangle_1$ for $\mathcal{S}_1$ and $\langle \cdot, \cdot \rangle_2$ for $\mathcal{S}_2$ is the linear map A^* such that

$$\forall x \in \mathcal{S}_1, y \in \mathcal{S}_2, \qquad \langle A(x), y \rangle_2 = \langle x, A^*(y) \rangle_1.$$

For our Jacobians:

$$\begin{aligned} \langle t_k, r_k \rangle_k &= \langle \partial f_k(t_{k-1}), r_k \rangle_k \\ &= \langle t_{k-1}, \partial f_k^*(r_k) \rangle_{k-1} \end{aligned}$$

so by defining the backward pass with $r_{k-1} = \partial f_k^*(r_k)$, we get

$$\langle t_k, r_k \rangle = \langle t_{k-1}, r_{k-1} \rangle$$

By induction, we get

$$\langle t_0, r_0 \rangle = \langle t_K, r_K \rangle$$

▸ **We are free to choose any scalar product $\langle \cdot, \cdot \rangle_k$. Does this choice affect the Jacobian values as well as gradient of f ?**

Matrix multiplication (Vectorized way)

Kronecker product properties:

$$\mathrm{vec}(AXB) = (B^\top \otimes A)\mathrm{vec}(X) \qquad (A \otimes B)^\top = A^\top \otimes B^\top$$

Deriving VJP and JVP for matrix product:

$$\begin{aligned} F(X) &= AX \\ G(\mathrm{vec}(X)) \triangleq \mathrm{vec}(F(X)) &= (I \otimes A)\mathrm{vec}(X) \\ J_G &= (I \otimes A) \\ J_G^\top \mathrm{vec}(R) &= (I \otimes A^\top)\mathrm{vec}(R) \\ \partial F^*[R] = \mathrm{mat}(J_G^\top \mathrm{vec}(R)) &= A^\top R \end{aligned}$$

▸ **How should we store the Jacobian in the forward pass to save it for the backward pass ?**

Matrix multiplication (Scalar product way)

For the scalar product

$$\langle X, Y \rangle = \sum_{i,j} X_{ij} Y_{ij} = \sum_{i,j} (X \odot Y)_{ij} = \langle \mathrm{vec}(X), \mathrm{vec}(Y) \rangle = \mathrm{tr}(XY^\top), \quad A^* = A^\top$$

Now, given a forward tangent T and a reverse tangent R

$$\langle AT, R \rangle = \langle T, A^\top R \rangle$$

so the backward pass computes $A^\top R$.

▸ **How to prove that $A^* = A^\top$?**

Inner products

$\langle\rangle_i$, the last one should be just product of scalars to use $r_K = 1$, the first one should be canonical to be one-hot. The inner ones can be anything. Multiplying it by 2 makes one adjoint be multiplied by 2, the other by 1/2, it just cancels out.

Broadcasting (Vectorized way)

Consider applying a scalar function f (e.g. $\tanh$ to each entry of a matrix X.)

$$
\begin{aligned}
(F(X))_{ij} &= f(X_{ij}) = f.(X) \\
G(\mathrm{vec}(X)) \triangleq \mathrm{vec}(F(X)) &= \mathrm{vec}(f.(X)) \\
J_G &= \mathrm{Diag}(\mathrm{vec}(f'.(X))) \\
J_G \mathrm{vec}(T) &= \mathrm{Diag}(\mathrm{vec}(f'.(X)))\mathrm{vec}(T) \\
\partial F[T] = \mathrm{mat}(J_G \mathrm{vec}(T)) &= f'.(X) \odot T \\
J_G^\top \mathrm{vec}(R) &= \mathrm{Diag}(\mathrm{vec}(f'.(X)))\mathrm{vec}(R) \\
\partial F^*[R] = \mathrm{mat}(J_G^\top \mathrm{vec}(R)) &= f'.(X) \odot R
\end{aligned}
$$

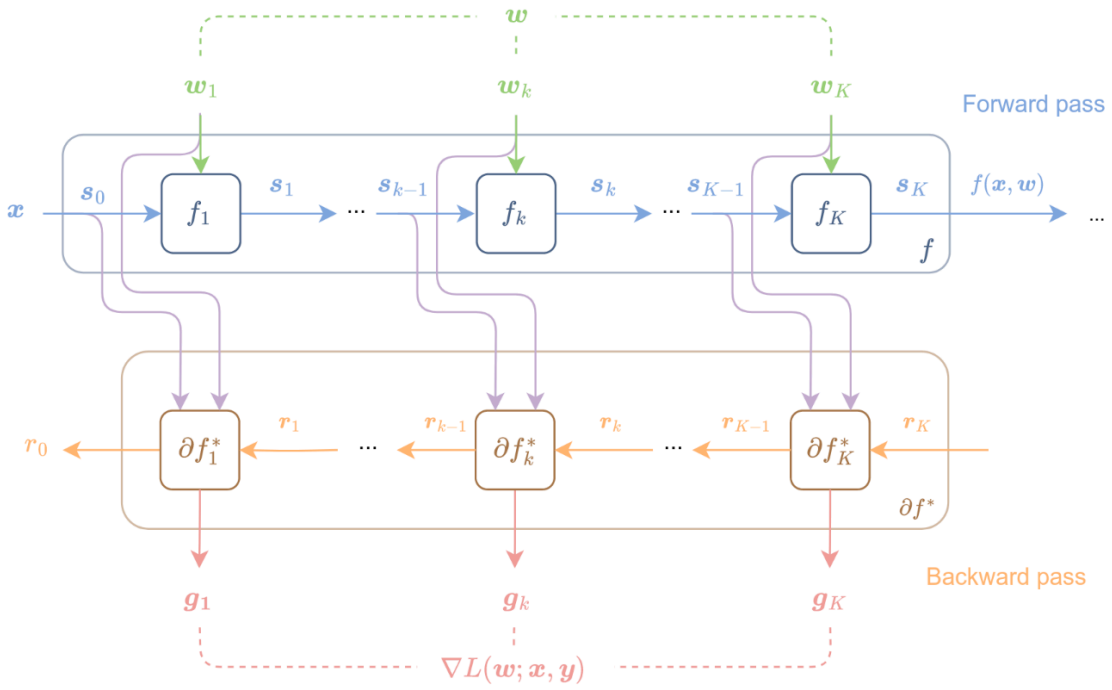
Broadcasting (Scalar product way)

$$\langle f'.(X) \odot T, R\rangle = \langle T, f'.(X) \odot R\rangle.$$

▸ **Let $A(X) = B \odot X$, what is the adjoint A^* ?**

▸ **What should be saved for the backward pass ?**

Putting everything together



Product of Jacobians

Suppose that we need to differentiate a composition of functions: $(f_n \circ f_{n-1} \circ \cdots \circ f_2 \circ f_1)(w)$. For each function, we can compute a jacobian given the value of its input. So, during a forward pass, we can compute all jacobians. We now just need to take the product of these jacobians:

$$J_n J_{n-1} \cdots J_2 J_1$$

While the product of matrices is associative, its computational complexity depends on the order of the multiplications! Let $d_i \times d_{i-1}$ be the dimension of J_i.

▶ **What is the complexity of forward mode**

▶ **What is the complexity of reverse mode**

▶ **What about the complexity of meeting in the middle between k and $k+1$?**

▶ **Which mode should be used depending on the d_i ?**

▶ **What about neural networks ?**

Wine example 🍷

```
data = DelimitedFiles.readdlm(
    last(save_image(URL("https://archive.ics.uci.edu/ml/machine-learning-databases/wine/wine.data"))),
    ',', Float64,
);
```

```
features =
13×178 Matrix{Float64}:
   14.23    13.2     13.16    14.37  …   13.71   13.4    13.27   13.17   14.13
    1.71     1.78     2.36     1.95       5.65    3.91    4.28    2.59    4.1
    2.43     2.14     2.67     2.5        2.45    2.48    2.26    2.37    2.74
   15.6     11.2     18.6     16.8       20.5    23.0    20.0    20.0    24.5
  127.0    100.0    101.0    113.0       95.0   102.0   120.0   120.0    96.0
    2.8      2.65     2.8      3.85  …    1.68    1.8     1.59    1.65    2.05
    3.06     2.76     3.24     3.49       0.61    0.75    0.69    0.68    0.76
    0.28     0.26     0.3      0.24       0.52    0.43    0.43    0.53    0.56
    2.29     1.28     2.81     2.18       1.06    1.41    1.35    1.46    1.35
    5.64     4.38     5.68     7.8        7.7     7.3    10.2     9.3     9.2
    1.04     1.05     1.03     0.86  …    0.64    0.7     0.59    0.6     0.61
    3.92     3.4      3.17     3.45       1.74    1.56    1.56    1.62    1.6
 1065.0   1050.0   1185.0   1480.0      740.0   750.0   835.0   840.0   560.0
```

```
features = permutedims(data[:, 2:end])
```

```
targets =
1×178 Matrix{Float64}:
 1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0  …  3.0  3.0  3.0  3.0  3.0  3.0  3.0  3.0
```

```
targets = permutedims(data[:, 1:1])
```

```
normalise (generic function with 1 method)
```

```
function normalise(x)
  μ = Statistics.mean(x, dims=2)
  σ = Statistics.std(x, dims=2, mean=μ)
  return (x .- μ) ./ σ
end
```

```
X = 13×178 Matrix{Float32}:
  1.51434    0.245597   0.196325    1.68679   …   0.331822   0.208643    1.39116
 -0.560668  -0.498009   0.0211715  -0.345835      1.73984    0.227053    1.57871
  0.2314    -0.825667   1.10621     0.486554     -0.38826    0.0126963   1.36137
 -1.1663    -2.48384   -0.267982   -0.806975      0.151234   0.151234    1.49872
  1.90852    0.018094   0.0881098   0.9283        1.41841    1.41841    -0.261969
  0.806722   0.567048   0.806722    2.48444   …  -1.12665   -1.03078    -0.391646
  1.03191    0.731565   1.21211     1.4624       -1.3408    -1.35081    -1.27072
 -0.657708  -0.818411  -0.497005   -0.979113      0.547563   1.35108     1.59213
  1.22144   -0.543189   2.12996     1.02925      -0.420888  -0.228701   -0.420888
  0.251009  -0.292496   0.268263    1.18273       2.21798    1.82976     1.78663
  0.361158   0.404908   0.317409   -0.426341  …  -1.60759   -1.56384    -1.52009
  1.84272    1.11032    0.786369    1.18074      -1.48127   -1.39676    -1.42493
  1.01016    0.962526   1.39122     2.32801       0.279786   0.295664   -0.593486
```

```
1 X = Float32.(normalise(features))
```

```
y =
1×178 Matrix{Float32}:
 -1.0  -1.0  -1.0  -1.0  -1.0  -1.0  -1.0  …  1.0  1.0  1.0  1.0  1.0  1.0  1.0  1.0
```

```
1 y = Float32.(targets .- 2)
```

Neural network

h = 16

```
W =
▸ [16×13 Matrix{Float32}:
   0.455791   0.192861   0.570114   0.800183   0.172319   …   0.312146   0.00240111   0.779776,
```

```
1 W = [rand(Float32, h, size(X, 1)), rand(Float32, size(y, 1), h)]
```

```
64.79143f0
```

```
1 sum((W[2] * tanh.(W[1] * X) - y).^2) / size(y, 2)
```

Forward mode

```
forward_pass (generic function with 1 method)
```

```
1 function forward_pass(W, X, y)
2     W1, W2 = W
3     y_1 = tanh.(W1 * X)
4     local_der_tanh = 1 .- y_1.^2
5     local_der_mse = 2 * (W2 * y_1 - y) / size(y, 2)
6     return local_der_tanh, local_der_mse
7 end
```

```
forward_diff (generic function with 1 method)
```

```
function forward_diff(W, X, y, j, k)
    W1, W2 = W
    T_1 = onehot(j, axes(W1, 1)) * onehot(k, axes(W1, 2))'
    J_1, J_2 = forward_pass(W, X, y)
    only((W2 * (J_1 .* (T_1 * X))) * J_2') # only: 1x1 matrix -> scalar
end
```

```
forward_diff (generic function with 2 methods)
```

```
function forward_diff(W, X, y)
    [forward_diff(W, X, y, i, j) for i in axes(W[1], 1), j in axes(W[1], 2)]
end
```

```
16×13 Matrix{Float32}:
  1.08394       0.379317     0.179821     …   0.00198301   0.567873    0.965956
 -0.000492529  -0.00385194  -0.000885715       0.00393592   0.00668501  0.00325014
  0.165919      0.0983145   -0.0557039        -0.0839093    0.310378    0.214846
  1.08466       0.203377     0.375327          0.476402     0.789234    1.13451
 -0.14793      -0.435628    -0.0616985         1.00439      1.27467     0.364239
  1.24543      -0.679034     1.0107       …    0.76879      0.628446    1.36107
  0.688892     -0.0403112    0.901926         -0.243525    -0.157552    0.867776
  ⋮                                      ⋱    ⋮
  0.294365      0.167444     0.499097     …  -0.171983    -0.132662    0.500498
  0.251275     -0.266478     0.213579          0.121441     0.377678    0.352928
  0.381609     -0.352307    -0.0722513         0.54867      0.865469    0.615858
  0.192073     -0.615239    -0.0493755         0.874016     1.11792     0.601956
  0.187852     -0.0608087    0.448009          0.207241     0.12643     0.39352
  0.718874     -0.127225    -0.018186     …   0.246473     0.0902219   0.827127
```

```
if h < 200 # Forward Diff start being too slow for 'h > 200'
    @time forward_diff(W, X, y)
end
```

```
  0.029018 seconds (6.24 k allocations: 12.209 MiB, 50.44% gc time)
```

Reverse mode

▶ **How to deduce the backward pass for reverse mode from the forward mode ?**

```
reverse_diff (generic function with 1 method)
```

```
function reverse_diff(W, X, y)
    J_1, J_2 = forward_pass(W, X, y)
    (J_1 .* (W[2]' * J_2)) * X'
end
```

```
16×13 Matrix{Float32}:
  1.08394       0.379317     0.179821     …   0.00198305   0.567874    0.965956
 -0.000492529  -0.00385194  -0.000885715       0.00393592   0.00668501  0.00325014
  0.165919      0.0983145   -0.0557039        -0.0839093    0.310378    0.214846
  1.08466       0.203377     0.375327          0.476402     0.789234    1.13451
 -0.14793      -0.435628    -0.0616985         1.00439      1.27467     0.364239
  1.24543      -0.679034     1.0107       …    0.76879      0.628446    1.36107
  0.688892     -0.0403111    0.901926         -0.243525    -0.157552    0.867776
  ⋮                                       ⋱   ⋮
  0.294365      0.167444     0.499097     …  -0.171983    -0.132662    0.500498
  0.251275     -0.266478     0.213579          0.121441     0.377678    0.352928
  0.381609     -0.352307    -0.0722513         0.54867      0.86547     0.615858
  0.192073     -0.615239    -0.0493755         0.874016     1.11792     0.601956
  0.187852     -0.0608087    0.448009          0.207241     0.12643     0.39352
  0.718874     -0.127225    -0.018186     …   0.246473     0.0902219   0.827127
```

```julia
@time reverse_diff(W, X, y)
```

```
  0.000090 seconds (25 allocations: 60.039 KiB)
```

🚀 GPU acceleration ⚡

```julia
if CUDA.functional()
    X_gpu = CUDA.CuArray(X)
    y_gpu = CUDA.CuArray(y)
    W_gpu = CUDA.CuArray.(W)
    @time reverse_diff(W_gpu, X_gpu, y_gpu)
end
```

▶ **Why is the GPU version slower than the CPU version ?**

Second-order

Consider a function $f : \mathbb{R}^n \to \mathbb{R}$, we want to compute the Hessian $\nabla^2 f(x)$, defined by

$$(\nabla^2 f(x))_{ij} = \frac{\partial^2 f}{\partial x_i \partial x_j}$$

Application: Given the optimization problem:

$$\begin{aligned} &\min_x f(x) \\ &\quad g_i(x) = 0 \quad \forall i \in \{1, \dots, m\} \end{aligned}$$

The Hessian of the Lagrangian $\mathcal{L}(x, \lambda) = f(x) - \lambda_1 g_1(x) - \cdots - \lambda_m g_m(x)$ is obtained as

$$\nabla^2_x \mathcal{L}(x, \lambda) = \nabla^2 f(x) - \sum_{i=1}^{m} \lambda_i \nabla^2 g_i(x)$$

Second-order AD

▸ **How can the Hessian of f be computed given an AD for Jacobian and gradient.**

▸ **Does the AD need to be the same for the gradient and the Jacobian ?**

Notation

- Let $f_k : \mathbb{R}^{d_{k-1}} \to \mathbb{R}^{d_k}$. $\partial f_k \triangleq \partial f_k(s_{k-1}) \in \mathbb{R}^{d_k \times d_{k-1}}$, $\partial^2 f_k \triangleq \partial^2 f_k(s_{k-1}) \in \mathbb{R}^{d_k \times d_{k-1} \times d_{k-1}}$ is a 3D array/tensor.
- Given $v \in \mathbb{R}^{d_{k-1}}$, by the product $(\partial^2 f_k \cdot v) \in \mathbb{R}^{d_k \times d_{k-1}}$, we denote the contraction of the 3rd (or 2nd since the tensor is symmetric over its last 2 dimensions) dimension:

$$(\partial^2 f_k \cdot v)_{ij} = \sum_{l=1}^{d_{k-1}} (\partial^2 f_k)_{ijl} \cdot v_l$$

- Given $u \in \mathbb{R}^{d_k}$, by the product $(u \cdot \partial^2 f_k) \in \mathbb{R}^{d_{k-1} \times d_{k-1}}$, we denote the contraction of the 1st dimension.

$$(u \cdot \partial^2 f_k)_{ij} = \sum_{l=1}^{d_k} u_l \cdot (\partial^2 f_k)_{lij}$$

- Both $\partial^2 f_k \cdot v$ and $u \cdot \partial^2 f_k$ are matrices so then we're back to matrix notations.

Chain rule

$$\begin{aligned}
\frac{\partial^2 (f_2 \circ f_1)}{\partial x_i \partial x_j} &= \frac{\partial}{\partial x_j} \left(\frac{\partial (f_2 \circ f_1)}{\partial x_i} \right) \\
&= \frac{\partial}{\partial x_j} \left(\partial f_2 \cdot \frac{\partial f_1}{\partial x_i} \right) \\
&= \left(\partial^2 f_2 \cdot \frac{\partial f_1}{\partial x_j} \right) \cdot \frac{\partial f_1}{\partial x_i} + \partial f_2 \cdot \frac{\partial^2 f_1}{\partial x_i \partial x_j}
\end{aligned}$$

In terms of the matrices $J_k = \partial f_k$ and $H_{kj} = \frac{\partial}{\partial x_j} J_k = \partial^2 f_k \cdot \frac{\partial s_{k-1}}{\partial x_j}$, it becomes

$$\frac{\partial^2 (f_2 \circ f_1)}{\partial x_i \partial x_j} = H_{2j} \cdot \frac{\partial f_1}{\partial x_i} + J_2 \cdot \frac{\partial^2 f_1}{\partial x_i \partial x_j}$$

Forward on forward

Given $\mathbf{Dual}(s_1, t_1)$ with $s_1 = \mathbf{Dual}(f_1(x), \frac{\partial f_1}{\partial x_j})$ and $t_1 = \mathbf{Dual}(\frac{\partial f_1}{\partial x_i}, \frac{\partial^2 f_1}{\partial x_i \partial x_j})$

1. Compute $s_2 = f_2(s_1) = (f_2(f_1(x)), J_2 \cdot \frac{\partial f_1}{\partial x_j}) = ((f_2 \circ f_1)(x), \partial (f_2 \circ f_1)/\partial x_j)$
2. Compute $J_{f_2}(s_1)$ which gives $\mathbf{Dual}(J_2, H_{2j})$
3. Compute

$$\begin{aligned}
J_{f_2}(s_1) \cdot t_1 &= \mathbf{Dual}(J_2, H_{2j}) \cdot \mathbf{Dual}(\frac{\partial f_1}{\partial x_i}, \frac{\partial^2 f_1}{\partial x_i \partial x_j}) \\
&= \mathbf{Dual}(J_2 \cdot \frac{\partial f_1}{\partial x_i}, J_2 \cdot \frac{\partial^2 f_1}{\partial x_i \partial x_j} + H_{2j} \cdot \frac{\partial f_1}{\partial x_i}) \\
&= \mathbf{Dual}(\frac{\partial (f_2 \circ f_1)}{\partial x_i}, \frac{\partial^2 (f_2 \circ f_1)}{\partial x_i \partial x_j})
\end{aligned}$$

▶ **What is the closed form expression for t_k in terms of the matrices J_k and H_{kj} ?**

Forward on reverse

Forward pass: Given $s_1 = \mathrm{Dual}(f_1(x), \frac{\partial f_1}{\partial x_j})$

1. Compute $s_2 = f_2(s_1)$ → same as forward on forward
2. Compute $J_{f_2}(s_1)$ → same as forward on forward

Reverse pass: Given $r_2 = \mathrm{Dual}((r_2)_1, (r_2)_2)$, compute

$$\begin{aligned} r_2 \cdot J_2 &= \mathrm{Dual}((r_2)_1, (r_2)_2) \cdot \mathrm{Dual}(J_2, H_{2j}) \cdot \\ &= \mathrm{Dual}((r_2)_1 \cdot J_2, (r_2)_2 \cdot J_2 + (r_2)_1 \cdot H_{2j}) \end{aligned}$$

▶ **Which value of r_k is solution for this recurrence equation ?**

▶ **What is the closed form expression for r_k in terms of the matrices J_k and H_{kj} ?**

Reverse on forward

Forward pass: Given $s_1 = \mathrm{Dual}(f_1(x), \frac{\partial f_1}{\partial x_i})$

1. Forward mode computes
   $s_2 = f_2(s_1) = (f_2(f_1(x)), J_2 \cdot \frac{\partial f_1}{\partial x_i}) = ((f_2 \circ f_1)(x), \partial(f_2 \circ f_1)/\partial x_i)$
2. The reverse mode computes the local Jacobian of this operation : $\partial s_2 / \partial s_1$. The local Jacobian of $(s_1)_1 \mapsto f_2((s_1)_1)$ is J_2. The local Jacobian of $s_1 \mapsto \partial f_2((s_1)_1)(s_1)_2$ is $(\partial^2 f_2((s_1)_1) \cdot (s_1)_2, \partial f_2((s_1)_1)) = (\partial^2 f_2(f_1(x)) \cdot \frac{\partial f_1}{\partial x_i}, \partial f_2(f_1(x)) = (H_{2i}, J_2)$

Reverse pass:

$$\begin{aligned} (r_1)_1 &= (r_2)_1 \cdot J_2 + (r_2)_2 \cdot H_{2i} \\ (r_1)_2 &= (r_2)_2 \cdot J_2 \end{aligned}$$

▶ **Which value of r_k is solution for this recurrence equation ?**

Reverse on reverse

Forward pass (2nd):

1. Forward pass computes $s_2 = f_2(s_1)$ → Jacobian $\partial s_2 / \partial s_1 = J_2$
2. Local Jacobian $J_2 = \partial f_2(s_1)$ → The Jacobian is the 3D array $\partial J_2 / \partial s_1 = \partial^2 f_2$
3. Backward pass computes $r_1 = r_2 \cdot \partial f_2(s_1)$ → Jacobian of $(s_1, r_2) \mapsto r_2 \cdot \partial f_2(s_1)$ is $(r_2 \cdot \partial^2 f_2(s_1), \partial f_2(s_1)) = (r_2 \cdot \partial^2 f_2, J_2)$. Note that here $r_2 \in \mathbb{R}^{d_k}$ is multiplying the first dimension of the tensor $\partial^2 f_2(s_1) \in \mathbb{R}^{d_k \times d_{k-1} \times d_{k-1}}$ so the result is a symmetric matrix of dimension $\mathbb{R}^{d_{k-1} \times d_{k-1}}$

Reverse pass (2nd): The result is r_0, let $\dot{r}_k$ be the second-order reverse tangent for r_k and $\dot{s}_k$ be the second-order reverse tangent of s_k. We have

$$\begin{aligned} \dot{r}_2 &= J_2 \cdot \dot{r}_1 \\ \dot{s}_1 &= (r_2 \cdot \partial^2 f_2(s_1)) \cdot \dot{r}_1 + \dot{s}_2 \cdot J_2 \end{aligned}$$

▶ **Which value of $\dot{s}_k, \dot{r}_k$ is solution for this recurrence equation ?**

▶ **What is the difference with reverse on forward and forward on reverse ?**

Acknowledgements and further readings

- `Dual` is inspired from ForwardDiff
- `Node` is inspired from micrograd
- Here is a good intro to AD
- Figures are from the The Elements of Differentiable Programming book

The End

Utils

```
using PlutoUI, PlutoUI.ExperimentalLayout, HypertextLiteral, PlutoTeachingTools, Statistics, CUDA, OneHotArrays
```

```
import PlotlyLight
```

```
import ComputationGraphExplorer as CGE
```

```
_latex (generic function with 1 method)
```

```
begin
    using LaTeXStrings
    import MathTeXEngine
    # Luxor asks Cairo for MathTeXEngine's fonts by name so we register them
    # to avoid having to install them on the system
    let fontconfig = Base.require(Base.PkgId(Base.UUID("a3f928ae-7b40-5064-980b-68af3947d34b"), "Fontconfig_jll"))
        add_font = Base.Libc.Libdl.dlsym(fontconfig.libfontconfig_handle, :FcConfigAppFontAddFile)
        dir = joinpath(pkgdir(MathTeXEngine), "assets", "fonts", "NewComputerModern")
        for file in readdir(dir; join = true)
            if endswith(file, ".otf")
                ccall(add_font, Cint, (Ptr{Cvoid}, Cstring), C_NULL, file)
            end
        end
    end
    # Luxor draws the italic glyphs of letters with the upright font so we
    # replace them with their Unicode mathematical italic counterpart.
    # The arguments of font commands such as `\mathrm{...}` are left untouched
    # and `\mathcal{X}` is replaced by the Unicode script letter since
    # MathTeXEngine would use the bold script one.
    _math_italic(c) = c == 'h' ? 'ℎ' : islowercase(c) ? Char(0x1D44E + (c - 'a')) : Char(0x1D434 + (c - 'A'))
    _math_script(c) = get(Dict('B' => 'ℬ', 'E' => 'ℰ', 'F' => 'ℱ', 'H' => 'ℋ', 'I' => 'ℐ', 'L' => 'ℒ', 'M' => 'ℳ', 'R' => 'ℛ'), c, Char(0x1D49C + (c - 'A')))
    function _latex(s)
        s = replace(s, r"\\mathcal\{([A-Z])\}" => m -> string(_math_script(m[end-1])))
        s = replace(s, r"\\(math[a-z]+|text[a-z]*|operatorname)\{[^}]*\}|\\[a-zA-Z]+|[a-zA-Z]" => m -> length(m) == 1 ? string(_math_italic(m[1])) : m)
        return latexstring(s)
    end
end
```

mathjax_figure

```
mathjax_figure(draw; width, height)
```

Luxor only draws the shapes, the labels are overlaid in HTML so that they are typeset by the MathJax of Pluto. `draw(label)` draws the shapes with the origin at the center and adds labels with `label(text, x, y; color, fontsize, class)`. Labels of class `equal` are centered between the labels of class `inner-product` once MathJax has set their width.

```
"""
    mathjax_figure(draw; width, height)

Luxor only draws the shapes, the labels are overlaid in HTML so that
they are typeset by the MathJax of Pluto.
`draw(label)` draws the shapes with the origin at the center and adds labels
with `label(text, x, y; color, fontsize, class)`.
Labels of class `equal` are centered between the labels of class
`inner-product` once MathJax has set their width.
"""
function mathjax_figure(draw; width, height)
    Luxor = CGE.Luxor
    labels = []
    function label(text, x, y; color = "black", fontsize = 14, class = "")
        push!(labels, @htl("""<span class="tex $class" style="position: absolute;
        left: $(x + width / 2)px; top: $(y + height / 2)px; transform:
        translate(-50%, -50%); white-space: nowrap; color: $color; font-size:
        $(fontsize)px;">\$$text\$</span>"""))
    end
    Luxor.Drawing(width, height, :svg)
    Luxor.origin()
    Luxor.background("white")
    draw(label)
    Luxor.finish()
    svg = Luxor.svgstring()
    @htl("""<div style="position: relative; width: $(width)px; height:
    $(height)px;">
    <div style="position: absolute; inset: 0;">$(HTML(svg))</div>
    $labels
    <script>
    // Center each `=` between the inner products once MathJax has set their width
    const root = currentScript.parentElement
    const products = [...root.querySelectorAll(".inner-product")]
    const equals = [...root.querySelectorAll(".equal")]
    const place = () => {
        const box = root.getBoundingClientRect()
        const scale = root.offsetWidth / box.width
```

```
        equals.forEach((equal, i) => {
            const left = products[i].getBoundingClientRect()
            const right = products[i + 1].getBoundingClientRect()
            equal.style.left = `${((left.right + right.left) / 2 - box.left) *
            scale}px`
        })
    }
    const observer = new ResizeObserver(place)
    products.forEach(product => observer.observe(product))
    invalidation.then(() => observer.disconnect())
    </script>
    </div>""")
end
```

chain_figure

```
chain_figure(annotate; width = 640, height = 360)
```

Draw the forward, tangent and reverse chains of $f_K \circ \cdots \circ f_1$ with `mathjax_figure`. Additional labels are added with `annotate(label, geometry)` where `label(text, x, y; color, fontsize, class)` adds a label and `geometry` gives the position of the columns `xs` with their index `ks`.

```
"""
    chain_figure(annotate; width = 640, height = 360)

Draw the forward, tangent and reverse chains of ``f_K \\circ \\cdots \\circ f_1``
with [`mathjax_figure`](@ref).
Additional labels are added with `annotate(label, geometry)` where
`label(text, x, y; color, fontsize, class)` adds a label and `geometry`
gives the position of the columns `xs` with their index `ks`.
"""
function chain_figure(annotate; width = 640, height = 360)
    mathjax_figure(; width, height) do _text
        Luxor = CGE.Luxor
        box_width = 45
        dy_up = 3box_width/8
        Δx = 90
        Δy = 90
        colors = (s = "#7ea6e0", t = "#67ab9f", r = "#ffb570")
        function _arrow(start, finish; rev, color)
            if rev
                start, finish = finish, start
            end
            Luxor.sethue(color)
            Luxor.arrow(start, finish; linewidth = 1.5, arrowheadlength = 10)
        end
        function draw_node(text, x, y; fill_color = "white", stroke_color =
        "#f79009", linewidth = 2, width = box_width, height = box_width)
            center = Luxor.Point(x, y)
            Luxor.sethue(fill_color)
            Luxor.box(center, width, height, 12, :fill)
            Luxor.sethue(stroke_color)
            Luxor.setline(linewidth)
            Luxor.box(center, width, height, 12, :stroke)
            _text(text, x, y)
        end
        function draw_line(f, v, y; rev, color = "black", stroke_color)
            dx_dots = box_width/4
            _text("$(v)_0", -3Δx, y; color)
```

```
            _arrow(Luxor.Point(-3Δx + box_width/2, y), Luxor.Point(-2Δx -
            box_width/2, y); rev, color)
            draw_node("$(f)_1", -2Δx, y; stroke_color)
            _arrow(Luxor.Point(-2Δx + box_width/2, y), Luxor.Point(-Δx - dx_dots,
            y); rev, color)
            _text("$(v)_1", -2Δx + box_width/2 + dy_up, y - dy_up; color)
            _text("\\cdots", -Δx, y)
            _arrow(Luxor.Point(-Δx + dx_dots, y), Luxor.Point(-box_width/2, y);
            rev, color)
            _text("$(v)_{k-1}", -box_width/2 - dy_up, y - dy_up; color)
            draw_node("$(f)_k", 0, y; stroke_color)
            _text("$(v)_k", box_width/2 + dy_up, y - dy_up; color)
            _arrow(Luxor.Point(box_width/2, y), Luxor.Point(Δx - dx_dots, y); rev,
            color)
            _text("\\cdots", Δx, y)
            _arrow(Luxor.Point(Δx + dx_dots, y), Luxor.Point(2Δx - box_width/2, y);
            rev, color)
            _text("$(v)_{K-1}", 2Δx - box_width/2 - dy_up, y - dy_up; color)
            draw_node("$(f)_K", 2Δx, y; stroke_color)
            _text("$(v)_K", 3Δx, y; color)
            _arrow(Luxor.Point(2Δx + box_width/2, y), Luxor.Point(3Δx -
            box_width/2, y); rev, color)
        end
        draw_line("f", "s", -Δy, rev = false, color = colors.s, stroke_color =
        "#546b8c")
        draw_line("\\partial f", "t", 0, rev = false; color = colors.t,
        stroke_color = "#40635c")
        draw_line("\\partial f^*", "r", Δy, rev = true; color = colors.r,
        stroke_color = "#b58b63")
        ks = ["0", "1", "k-1", "k", "K-1", "K"]
        xs = [-3Δx, -2Δx + box_width/2 + dy_up, -box_width/2 - dy_up, box_width/2 +
        dy_up, 2Δx - box_width/2 - dy_up, 3Δx]
        annotate(_text, (; ks, xs, Δy, dy_up, colors))
    end
end
```

fanout_figure

```
fanout_figure(; dup, width = 640, height = 260)
```

Draw the forward and reverse passes for `x` used by both `f_1` and `f_2` with `mathjax_figure`. If `dup` is `true`, `x` is first duplicated with `dup` so that we get a chain.

```
"""
    fanout_figure(; dup, width = 640, height = 260)

Draw the forward and reverse passes for `x` used by both `f_1` and `f_2`
with [`mathjax_figure`](@ref).
If `dup` is `true`, `x` is first duplicated with `dup` so that we get a chain.
"""
function fanout_figure(; dup, width = 640, height = 260)
    mathjax_figure(; width, height) do label
        Luxor = CGE.Luxor
        box_width = 45
        colors = (s = "#7ea6e0", r = "#ffb570", g = "gray")
        function node(text, center; width = box_width)
            Luxor.sethue("white")
            Luxor.box(center, width, box_width, 12, :fill)
            Luxor.sethue("#546b8c")
            Luxor.setline(2)
            Luxor.box(center, width, box_width, 12, :stroke)
            label(text, center.x, center.y)
        end
        # Forward arrow from `p` to `q` and the reverse arrow next to it on
        # the side given by the sign of `side`
        function arrows(p, q; side = 1)
            d = (q - p) / Luxor.distance(p, q)
            n = side * Luxor.Point(d.y, -d.x)
            Luxor.sethue(colors.s)
            Luxor.arrow(p, q; linewidth = 1.5, arrowheadlength = 10)
            Luxor.sethue(colors.r)
            Luxor.arrow(q + 8n, p + 8n; linewidth = 1.5, arrowheadlength = 10)
            return (p + q) / 2, n
        end
        x_f, x_s, Δy = 40, 230, 60
        x_x = dup ? -280 : -180
        label("x", x_x, 0; color = colors.s)
        label("r_0 = {\\color{red}?}", x_x, 28; color = colors.r)
        start = Luxor.Point(x_x + 14, 0)
        if dup
            x_dup = -180
            Luxor.sethue(colors.g)
            Luxor.setline(1)
```

```
            Luxor.setdash("dash")
            Luxor.box(Luxor.Point(x_dup, 0), 80, 70, 8, :stroke)
            Luxor.box(Luxor.Point(x_f, 0), 80, 200, 8, :stroke)
            Luxor.setdash("solid")
            label("g_1", x_dup, -47; color = colors.g)
            label("g_2", x_f, -112; color = colors.g)
            arrows(start, Luxor.Point(x_dup - 25, 0); side = -1)
            node("\\text{dup}", Luxor.Point(x_dup, 0); width = 50)
            start = Luxor.Point(x_dup + 25, 0)
        end
        for (i, side) in [(1, 1), (2, -1)]
            y = (2i - 3) * Δy
            mid, n = arrows(start, Luxor.Point(x_f - box_width / 2, y); side)
            if dup
                p = mid - 12n
                label("x", p.x, p.y; color = colors.s)
            end
            p = mid + 24n
            label("\\partial f_$(i)^* r_$i", p.x, p.y; color = colors.r, fontsize =
            12)
            node("f_$i", Luxor.Point(x_f, y))
            _, n = arrows(Luxor.Point(x_f + box_width / 2, y), Luxor.Point(x_s -
            20, y); side)
            label("s_$i", x_s, y; color = colors.s)
            p = Luxor.Point(x_s, y) + 22n
            label("r_$i", p.x, p.y; color = colors.r)
        end
    end
end
```

```
import LinearAlgebra, DelimitedFiles
```

```
img (generic function with 3 methods)
```

```
qa (generic function with 2 methods)
```